

PRISM Analytics — Interview Masterclass Notes

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■ Section 1: PRISM Retail Analytics (Project 1 Workflow)

Step-by-Step Making:

- 1. **Star Schema Database** (database/schema.sql): Fact table order_items (sales, profit, quantity, discount) linked via Foreign Keys to dimension tables customers, products, and orders. Created B-Tree indexes on order_date and customer_id for fast joins.
- 2. **Synthetic Data Generator** (database/seed_data.py): Seeded 11,000+ realistic transaction line items spanning 2 years.
- 3. **SQL Analytics Engine** (queries.py): Engineered KPIs, monthly revenue trends via date truncation strftime('%Y-%m'), and Customer Lifetime Value (CLV).
- 4. **RFM Customer Segmentation**: Applied window functions NTILE(4) OVER (...) to bucket customers into Recency, Frequency, and Monetary quartiles (Champions, Loyal, At Risk).
- 5. **Flask REST Backend** (app.py): Implemented connection pooling per request and safe SQL playground regex protection (blocking DROP, DELETE, UPDATE).
- 6. **Interactive Frontend** (public/): Custom dark CSS Grid UI, Vanilla JS fetchers, and Chart.js graphics.

■ Section 2: PRISM Gold Analytics (Project 2 Workflow)

Indian Retail Gold Rate Calculation Formula:

- **Troy Ounce to Grams**: 1 Troy Ounce = **31.1035 grams**.
- **Raw Spot Price (INR/g)**: (Spot USD per oz / 31.1035) * 83.50 (USD/INR Exchange Rate)
- **Indian Retail Multiplier (1.30)**: Includes 15% Customs Duty + 3% GST + 12% Bullion Import Margin & Jeweler baseline = **1.30 Factor** (~₹15,400 / 1g for 24K gold).
- **Purity Scaling**: **24K** (100% = x1.00), **22K** (91.67% = x0.9167), **18K** (75.00% = x0.7500).
- **Time-Series SQL Window Functions**: Calculated 7-day rolling moving averages using:
AVG(price_inr_24k_1g) OVER (ORDER BY date ROWS BETWEEN 6 PRECEDING AND CURRENT ROW)

■ Section 3: End-to-End Data Workflow (Life of a Request)

- 1. **Client Interaction**: User opens http://localhost:5000 in browser.
- 2. **Static Asset Delivery**: Flask serves index.html, CSS design tokens, and script.js.
- 3. **Asynchronous Fetch Calls**: JS fires parallel requests (e.g. GET /api/kpis, GET /api/revenue-trend).
- 4. **Database Query Execution**: Flask opens SQLite connection via get_conn(), executes parameterized SQL in queries.py.
- 5. **Serialization & Response**: SQLite tuples are formatted into JSON list of dicts and returned to browser.
- 6. **UI Rendering**: JS receives JSON payload and updates DOM metrics cards and renders **Chart.js** canvas elements.

■ Section 4: Top Interview Questions & Flashcard Answers

Interview Question	Bulletproof Technical Answer
Why Star Schema over 3NF?	Star Schema optimizes for OLAP (analytics) by flattening dimensions, reducing multi-join latency, and speeding up SUM/AVG aggregations.
How do you prevent SQL Injection in open playgrounds?	Enforce strict regex filtering that rejects DML/DDL mutation keywords (DROP, DELETE, UPDATE) and requires queries to start with SELECT or WITH.
What is the role of JULIANDAY in SQLite?	Computes date differences in fractional days (e.g. JULIANDAY('2024-12-31') - JULIANDAY(MAX(order_date))) for Recency calculation).
What is NTILE(4)?	A SQL window function that divides an ordered dataset into 4 equal quartiles (1 = Top 25%) without reducing row counts.
Why SQLite instead of PostgreSQL?	SQLite offers zero-network latency in-process reading for lightweight single-server setups. Queries can easily scale to PostgreSQL.